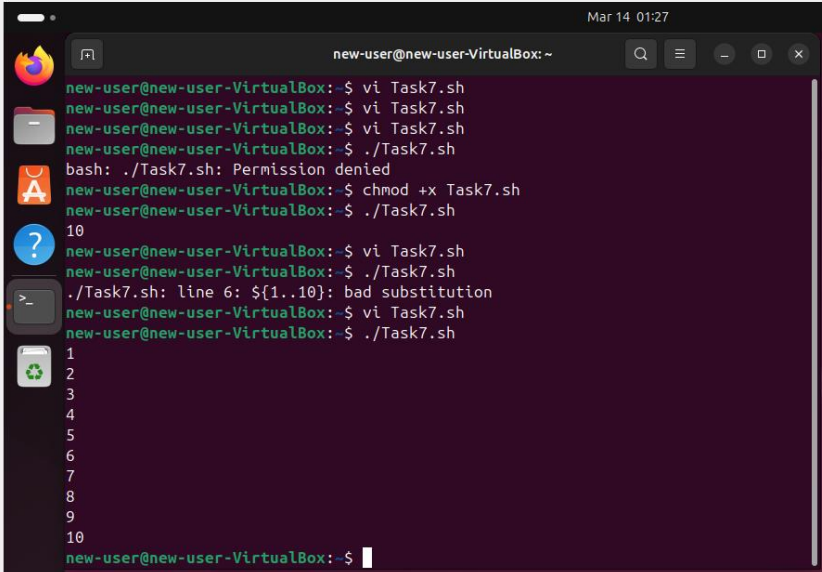


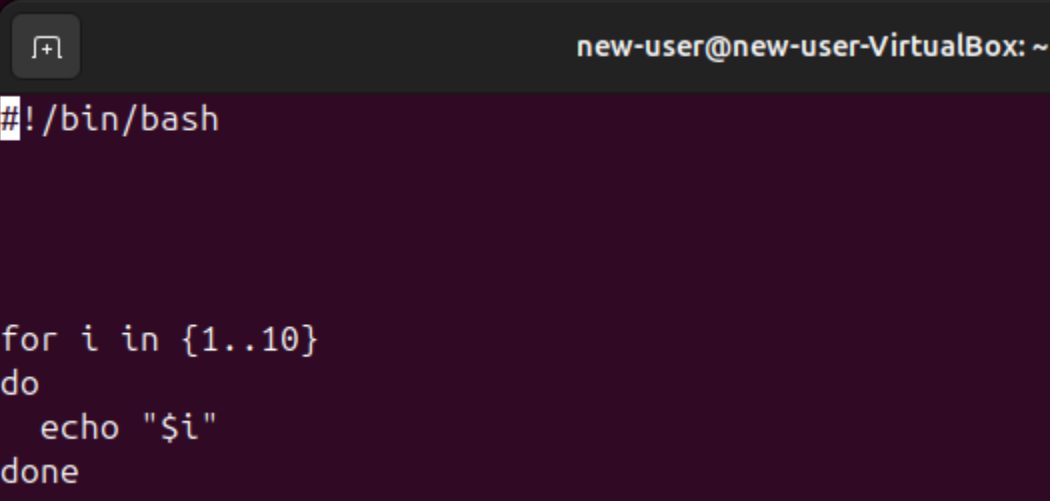
OS LAB 3 B TASKS

BAZIL UDDIN KHAN 24K-0559 BSCS4H

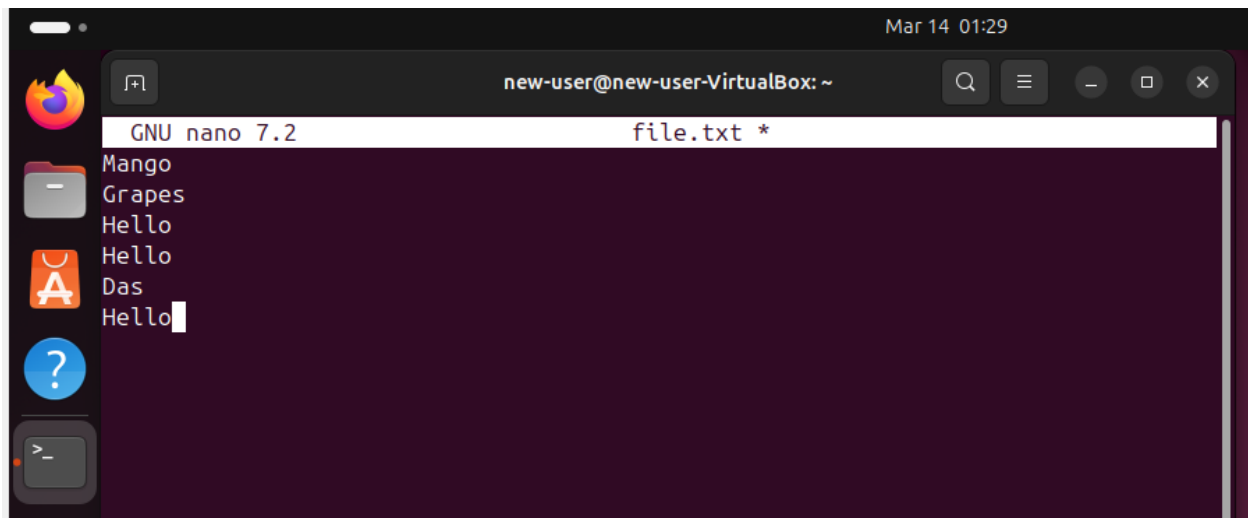
TASK#6).



```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox: $ vi Task7.sh  
new-user@new-user-VirtualBox: $ vi Task7.sh  
new-user@new-user-VirtualBox: $ vi Task7.sh  
new-user@new-user-VirtualBox: $ ./Task7.sh  
bash: ./Task7.sh: Permission denied  
new-user@new-user-VirtualBox: $ chmod +x Task7.sh  
new-user@new-user-VirtualBox: $ ./Task7.sh  
10  
new-user@new-user-VirtualBox: $ vi Task7.sh  
new-user@new-user-VirtualBox: $ ./Task7.sh  
./Task7.sh: line 6: ${1..10}: bad substitution  
new-user@new-user-VirtualBox: $ vi Task7.sh  
new-user@new-user-VirtualBox: $ ./Task7.sh  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
new-user@new-user-VirtualBox: $
```

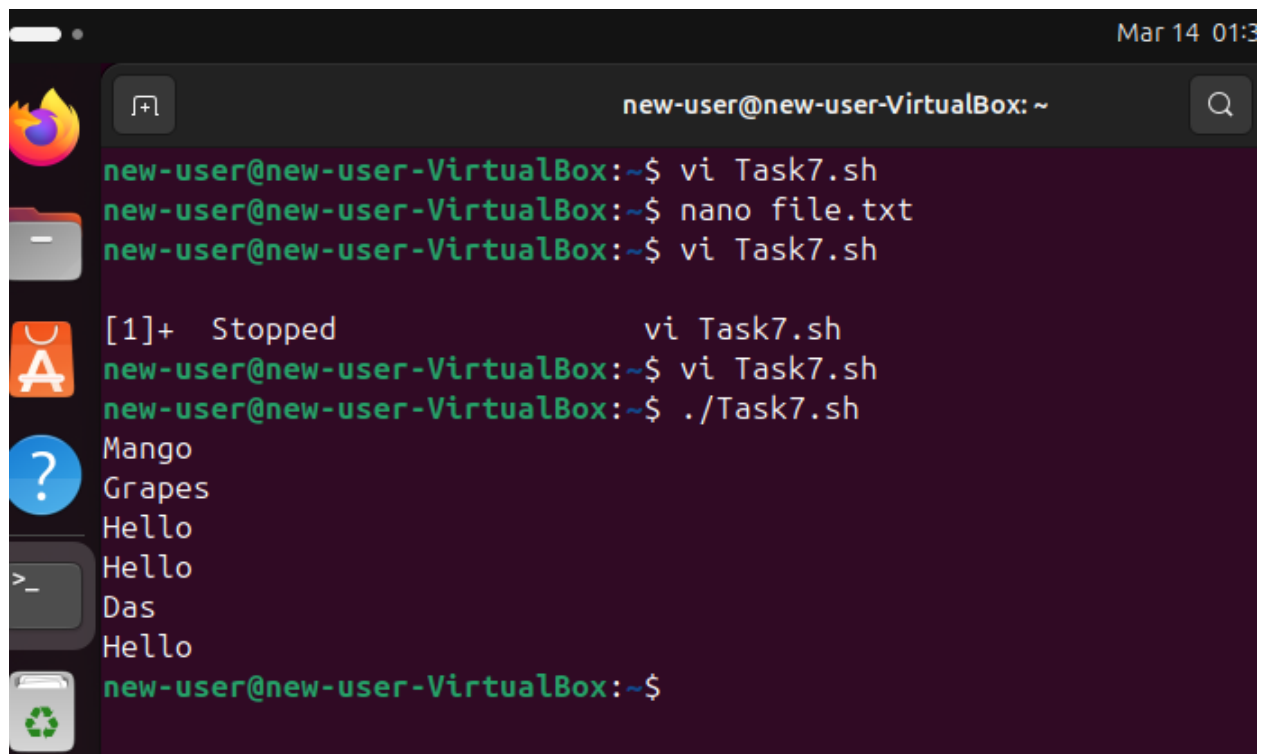


```
new-user@new-user-VirtualBox: ~  
#!/bin/bash  
  
for i in {1..10}  
do  
    echo "$i"  
done
```



A terminal window titled "new-user@new-user-VirtualBox: ~" with a timestamp of "Mar 14 01:29". The window shows the GNU nano 7.2 editor editing a file named "file.txt". The file contains the following text:

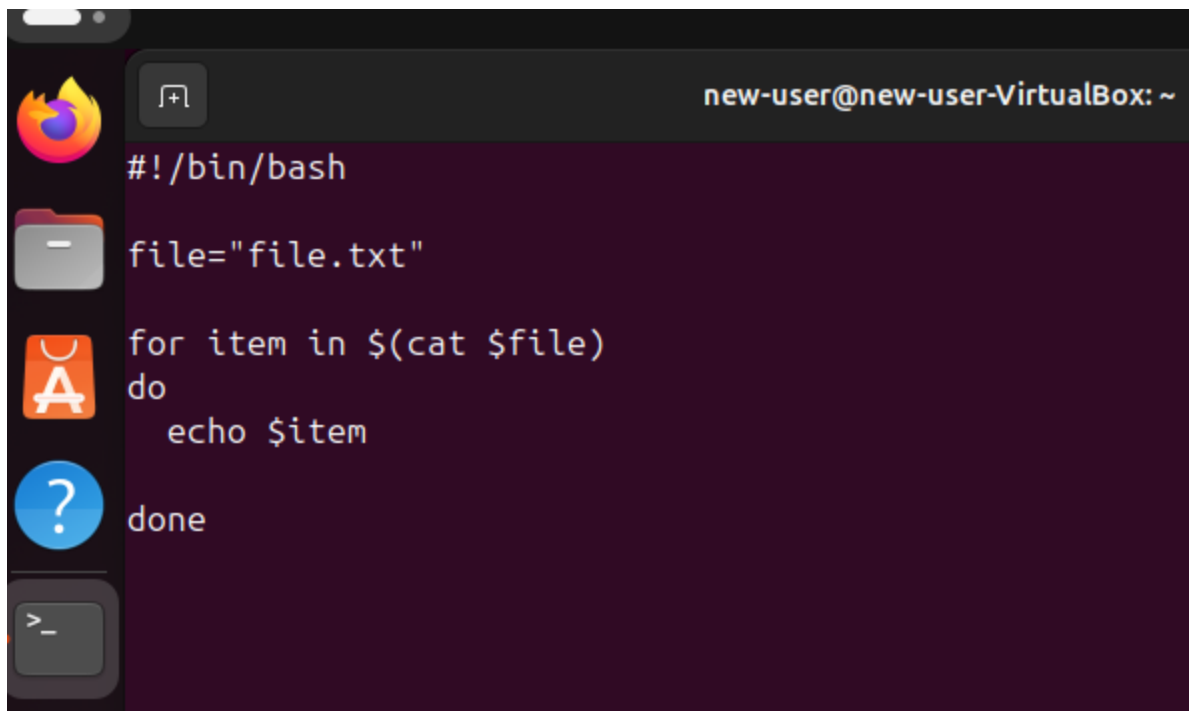
```
Mango
Grapes
Hello
Hello
Das
Hello
```



A terminal window titled "new-user@new-user-VirtualBox: ~" with a timestamp of "Mar 14 01:3". The window shows a series of commands and their outputs:

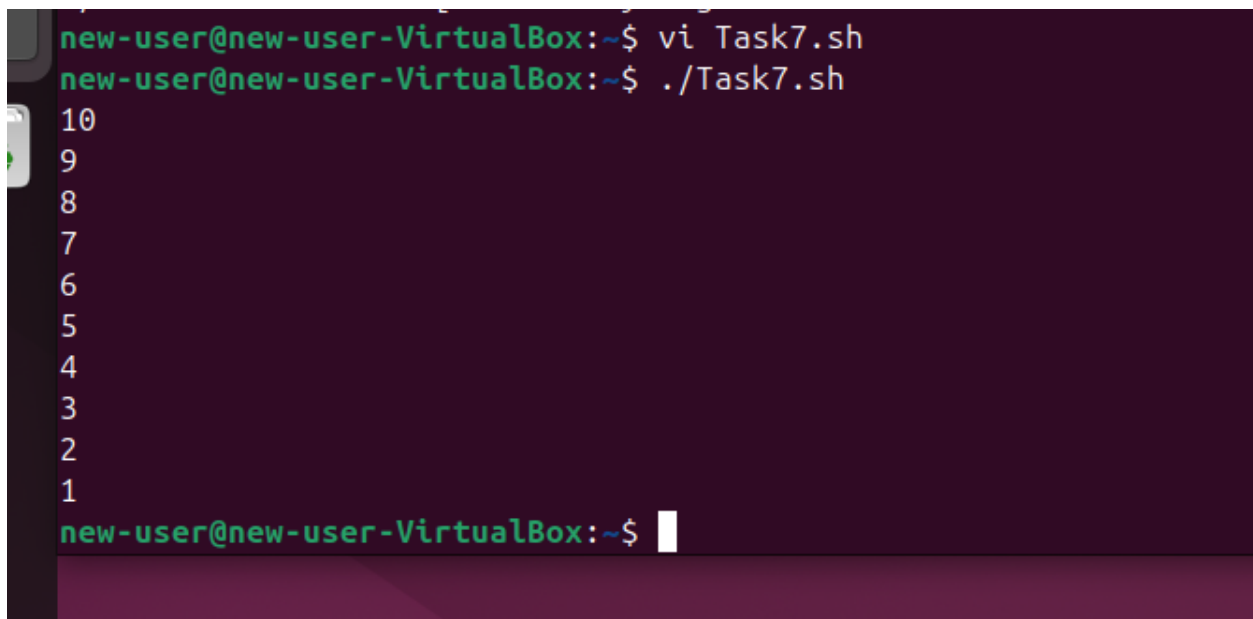
```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ nano file.txt
new-user@new-user-VirtualBox:~$ vi Task7.sh

[1]+  Stopped                  vi Task7.sh
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
Mango
Grapes
Hello
Hello
Das
Hello
new-user@new-user-VirtualBox:~$
```



A terminal window titled "new-user@new-user-VirtualBox: ~". The terminal shows a shell script being executed. The script starts with a shebang line, followed by a variable assignment, a for loop, and a done statement. On the left side of the terminal, there is a vertical sidebar with four icons: a Firefox logo, a folder icon, an application icon, and a question mark icon.

```
#!/bin/bash
file="file.txt"
for item in $(cat $file)
do
    echo $item
done
```



A terminal window showing the execution of a script. The prompt is "new-user@new-user-VirtualBox: ~\$". The user enters "vi Task7.sh" and then ". /Task7.sh". The terminal displays line numbers 1 through 10 on the left side. The prompt "new-user@new-user-VirtualBox: ~\$" is shown at the bottom with a cursor.

```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
10
9
8
7
6
5
4
3
2
1
new-user@new-user-VirtualBox:~$
```

```
new-user@new-user-virtualBox:~$ vi Task7.sh
#!/bin/bash

num=10

while [ $num -gt 0 ]
do
    echo $num

    let num--
done
```

```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
1
2
3
4
5
new-user@new-user-VirtualBox:~$
```

```
#!/bin/bash

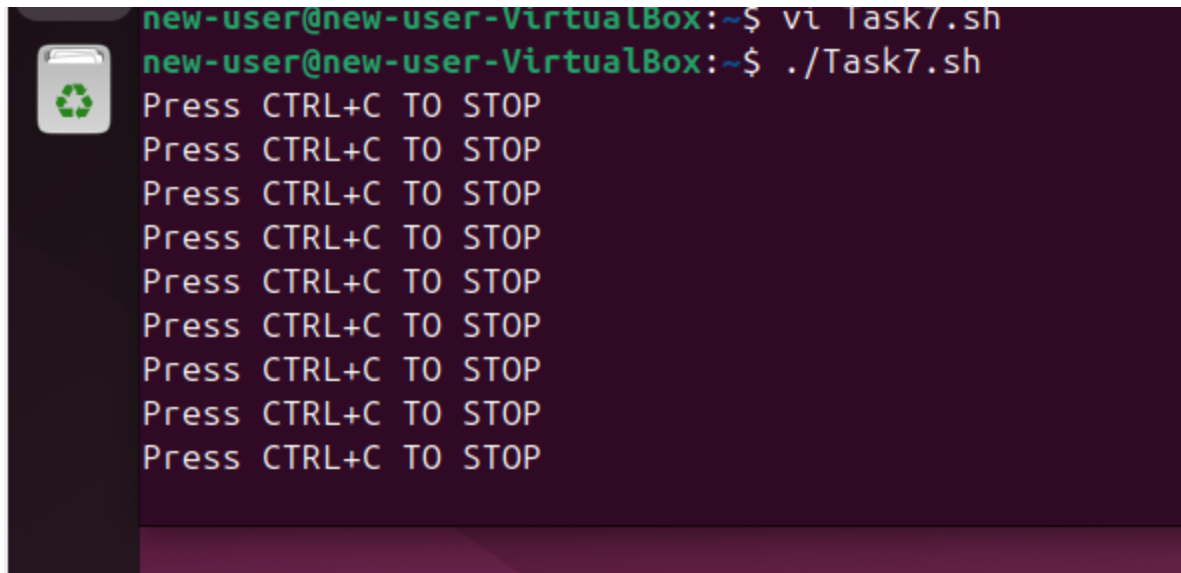
i=1
until [ $i -gt 5 ]
do
    echo $i

    let i++

done
```

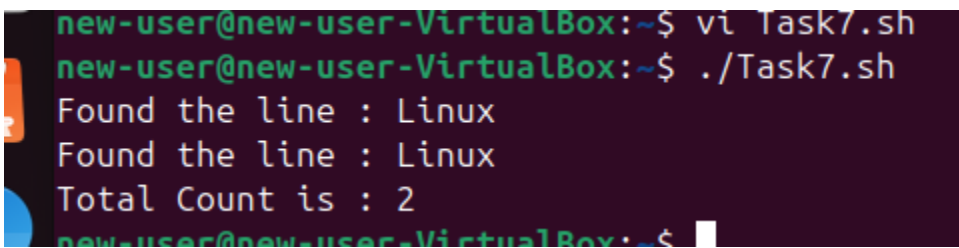
```
#!/bin/bash

while true
do
    echo "Press CTRL+C TO STOP "
    sleep 2
done
```



```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
```

TASK#7).



```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
Found the line : Linux
Found the line : Linux
Total Count is : 2
new-user@new-user-VirtualBox:~$
```

```
#!/bin/bash
```

```
file=file.txt
```

```
count=0
```

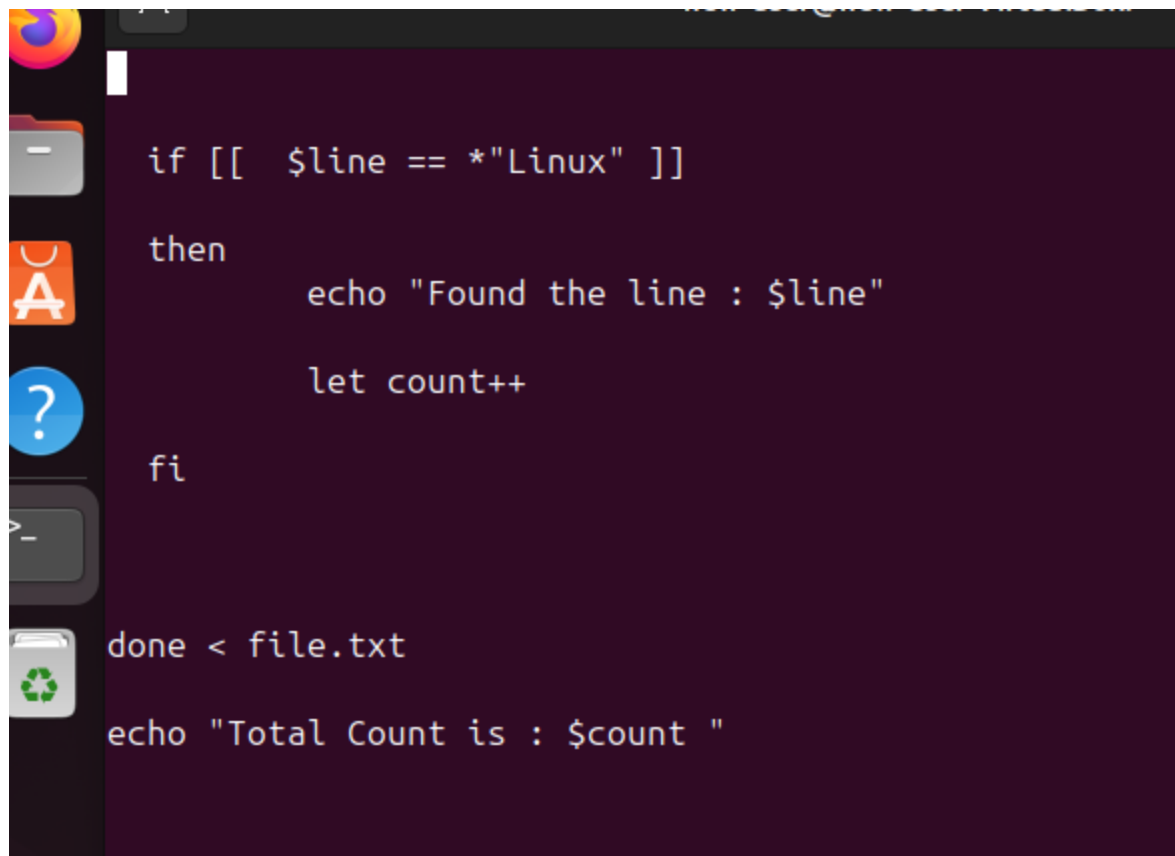
```
while read line
```

```
do
```

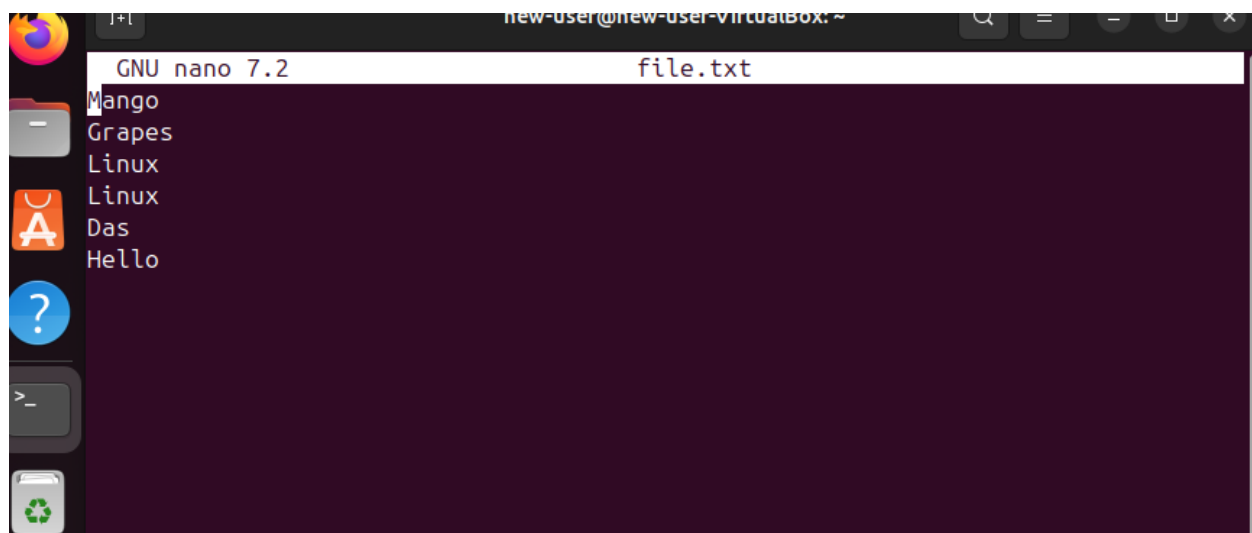
```
if [[ $line == *"Linux" ]]
```

```
then
```

```
    echo "Found the line : $line"
```

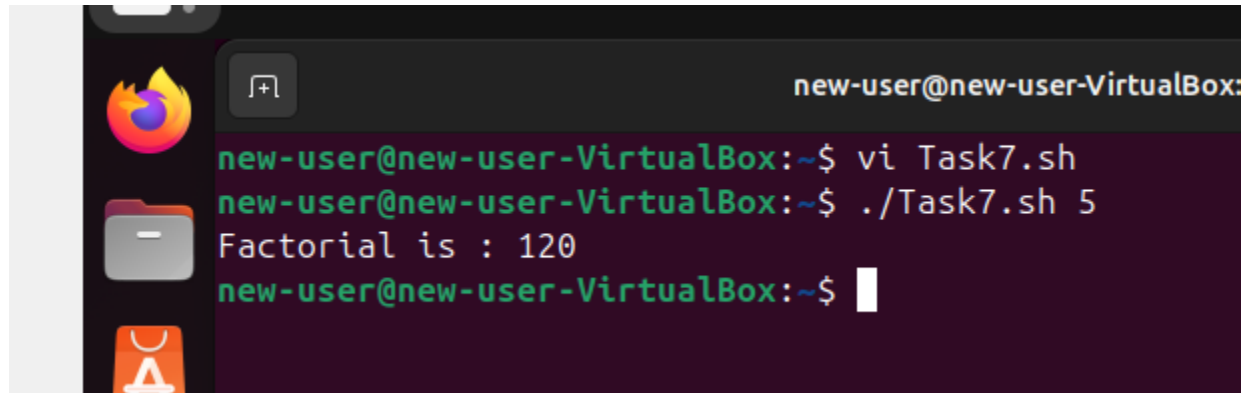


```
if [[ $line == *"Linux" ]]
then
    echo "Found the line : $line"
    let count++
fi
done < file.txt
echo "Total Count is : $count "
```

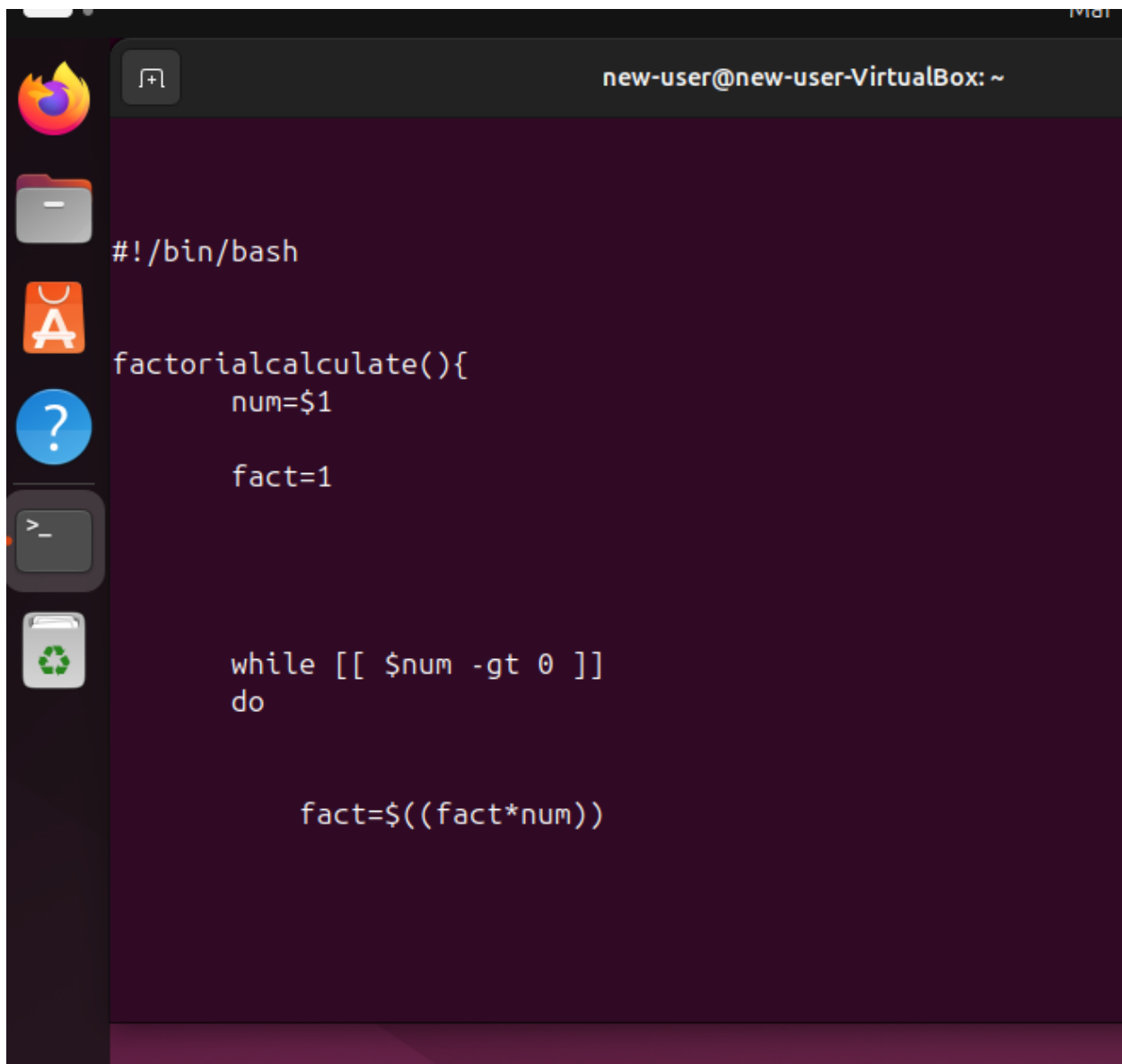


```
GNU nano 7.2 file.txt
Mango
Grapes
Linux
Linux
Das
Hello
```


TASK#8).

A terminal window titled 'new-user@new-user-VirtualBox:'. The window has a dark background with green text. On the left side, there is a vertical bar with three icons: a Firefox logo, a folder icon, and an Ubuntu logo. The terminal shows the following commands and output:

```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh 5
Factorial is : 120
new-user@new-user-VirtualBox:~$
```



A terminal window titled "new-user@new-user-VirtualBox: ~" with a dark purple background. On the left side, there is a vertical dock with icons for Firefox, a file manager, an application store, a help icon, a terminal, and a trash can. The terminal text is as follows:

```
#!/bin/bash

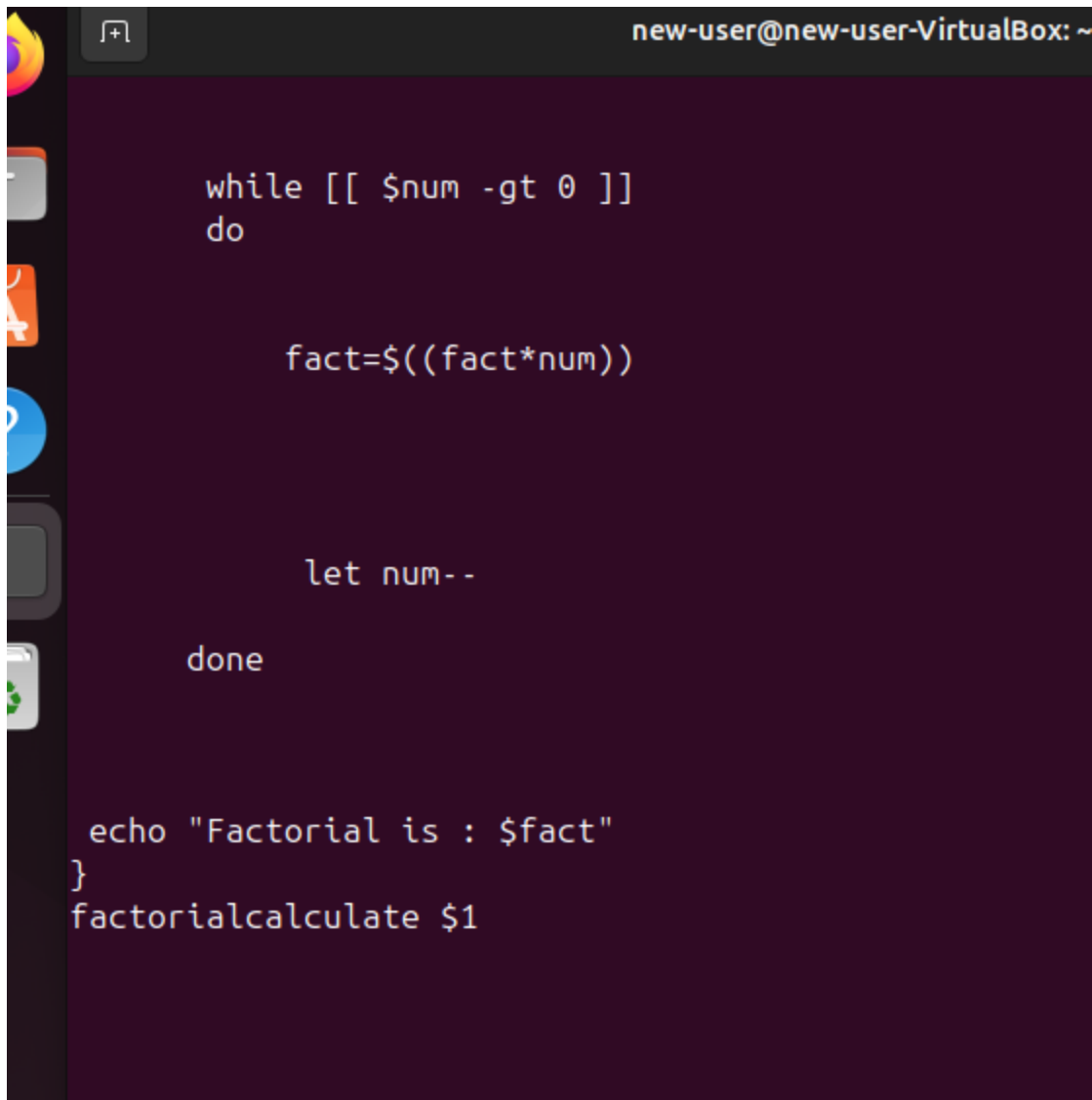
factorialcalculate(){
    num=$1

    fact=1

    while [[ $num -gt 0 ]]
    do

        fact=$((fact*num))

    done
}
```



A terminal window titled "new-user@new-user-VirtualBox: ~" with a dark purple background. The window contains a shell script for calculating the factorial of a number. The script uses a while loop to multiply the current number by the factorial of the previous number, then decrements the number until it reaches zero. The final result is printed using echo. The script is as follows:

```
while [[ $num -gt 0 ]]
do

    fact=$((fact*num))

    let num--

done

echo "Factorial is : $fact"
}
factorialcalculate $1
```

TASK#9).

```
new-user@new-user-VirtualBox:~$ chmod +x rename_files.sh
new-user@new-user-VirtualBox:~$ ./rename_files.sh . file
Starting file renaming in directory: .
New naming pattern: fileN.ext
Renamed: a.txt -> file1.txt
Renamed: b.txt -> file2.txt
Renamed: c.txt -> file3.txt
Renamed: file1. -> file4.
```

```
Renamed: file10. -> file5.  
Renamed: file11. -> file6.  
Renamed: file12. -> file7.  
Renamed: file13. -> file8.  
Renamed: file14. -> file9.  
Renamed: file15. -> file10.  
Renamed: file16. -> file11.  
Renamed: file17. -> file12.  
Renamed: file18. -> file13.  
Renamed: file19. -> file14.  
Renamed: file2. -> file15.  
Renamed: file20. -> file16.  
Renamed: file21. -> file17.  
Renamed: file22. -> file18.  
Renamed: file23. -> file19.  
Renamed: file24. -> file20.  
Renamed: file25. -> file21.  
Renamed: file26. -> file22.  
Renamed: file27 -> file23  
Renamed: file27. -> file24.  
Renamed: file28. -> file25.  
Renamed: file29. -> file26.  
Renamed: file3. -> file27.  
Renamed: file30. -> file28.  
Renamed: file31. -> file29.  
Renamed: file32. -> file30.  
Renamed: file33. -> file31.  
Renamed: file34. -> file32.  
Renamed: file35. -> file33.  
Renamed: file36. -> file34.  
Renamed: file4. -> file35.  
Renamed: file5. -> file36.  
Renamed: file6. -> file37.
```

```
Renamed: file2. -> file15.  
Renamed: file20. -> file16.  
Renamed: file21. -> file17.  
Renamed: file22. -> file18.  
Renamed: file23. -> file19.  
Renamed: file24. -> file20.  
Renamed: file25. -> file21.  
Renamed: file26. -> file22.  
Renamed: file27 -> file23  
Renamed: file27. -> file24.  
Renamed: file28. -> file25.  
Renamed: file29. -> file26.  
Renamed: file3. -> file27.  
Renamed: file30. -> file28.  
Renamed: file31. -> file29.  
Renamed: file32. -> file30.  
Renamed: file33. -> file31.  
Renamed: file34. -> file32.  
Renamed: file35. -> file33.  
Renamed: file36. -> file34.  
Renamed: file4. -> file35.  
Renamed: file5. -> file36.  
Renamed: file6. -> file37.  
Renamed: file7. -> file38.  
Renamed: file8. -> file39.  
Renamed: file9. -> file40.  
Renamed: rename_files.sh -> file41.sh  
Renaming complete.  
Total files processed: 41  
Errors encountered: 0
```

I had different files in my folder but first when I runned my code it just converted the name but the file extension was not conserved so that's why there are files like file5. Etc.

```
#!/bin/bash
```

```
if [ "$#" -ne 2 ]; then  
    echo "Usage: $0 <directory_path> <new_file_name_pattern>"  
    exit 1  
fi
```

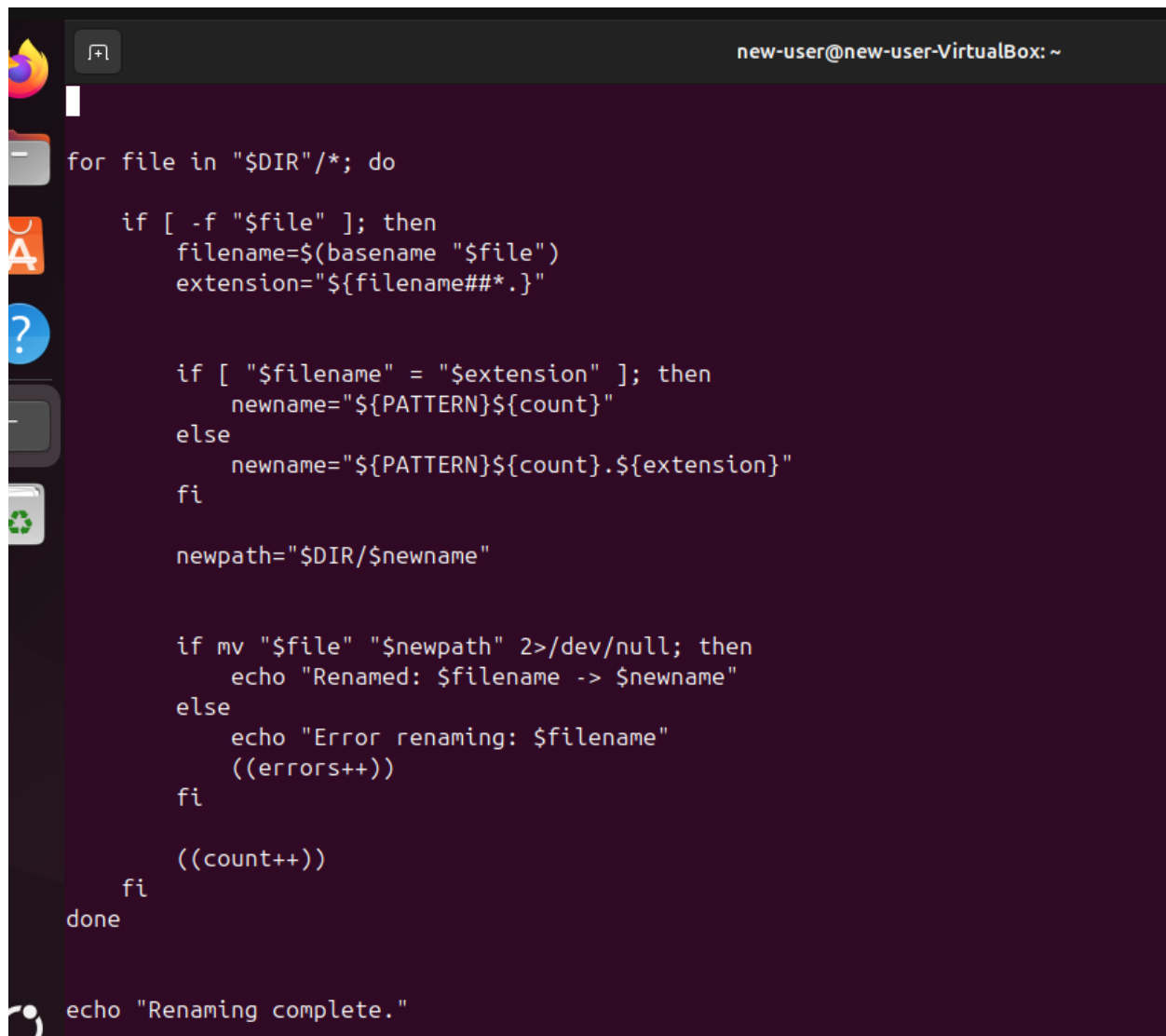
```
DIR="$1"  
PATTERN="$2"
```

```
if [ ! -d "$DIR" ]; then  
    echo "Error: Directory '$DIR' does not exist."  
    exit 1  
fi
```

```
count=1  
errors=0
```

```
echo "Starting file renaming in directory: $DIR"  
echo "New naming pattern: ${PATTERN}N.ext"
```

```
for file in "$DIR"/*; do  
    if [ -f "$file" ]; then  
        filename=$(basename "$file")  
        extension="${filename##*.}"
```



A terminal window titled "new-user@new-user-VirtualBox: ~" with a dark purple background. On the left, there is a vertical sidebar with icons for a web browser, a file manager, a terminal, a help icon, and a trash can. The terminal displays a shell script for renaming files. The script uses a loop to iterate over files in a directory, checks if they are regular files, extracts their base name and extension, and then renames them with a counter. It also includes error handling and a final completion message.

```
for file in "$DIR"/*; do

    if [ -f "$file" ]; then
        filename=$(basename "$file")
        extension="${filename##*}."

        if [ "$filename" = "$extension" ]; then
            newname="${PATTERN}${count}"
        else
            newname="${PATTERN}${count}.${extension}"
        fi

        newpath="$DIR/$newname"

        if mv "$file" "$newpath" 2>/dev/null; then
            echo "Renamed: $filename -> $newname"
        else
            echo "Error renaming: $filename"
            ((errors++))
        fi

        ((count++))
    fi
done

echo "Renaming complete."
```




```
if [ -f "$file" ]; then
    filename=$(basename "$file")
    extension="${filename##*.}"

    if [ "$filename" = "$extension" ]; then
        newname="${PATTERN}${count}"
    else
        newname="${PATTERN}${count}.${extension}"
    fi

    newpath="$DIR/$newname"

    if mv "$file" "$newpath" 2>/dev/null; then
        echo "Renamed: $filename -> $newname"
    else
        echo "Error renaming: $filename"
        ((errors++))
    fi

    ((count++))
fi
done

echo "Renaming complete."
echo "Total files processed: $((count-1))"
echo "Errors encountered: $errors"
```

TASK#10).

```
new user@new user virtua
#!/bin/bash

if [ "$#" -ne 2 ]; then

    echo " Usage: $0 <directory_path> <days_old>"

    exit 1
fi

DIR="$1"
DAYS="$2"

if [ ! -d "$DIR" ]; then

    echo " Error: Directory '$DIR' does not exist."
    exit 1
fi

echo " Starting cleanup in: $DIR"

echo " Removing files older than $DAYS days..."

files_before=$(find "$DIR" -type f -mtime +"$DAYS" | wc -l)
```

```
new-user@new-user-VirtualBox: ~  
  
    echo " Error: Directory '$DIR' does not exist."  
    exit 1  
fi  
  
echo " Starting cleanup in: $DIR"  
echo " Removing files older than $DAYS days..."  
  
files_before=$(find "$DIR" -type f -mtime +"$DAYS" | wc -l)  
  
find "$DIR" -type f -mtime +"$DAYS" -exec rm -f {} \;  
  
dirs_before=$(find "$DIR" -type d -empty | wc -l)  
find "$DIR" -type d -empty -delete  
  
echo " Cleanup complete!"  
  
echo " Files removed: $files_before"  
  
echo " Empty directories removed: $dirs_before"  
  
file12. file16. file1.txt file23 file27. file30. file34.  
new-user@new-user-VirtualBox:~$ ./cleanup.sh . 7  
Starting cleanup in: .  
Removing files older than 7 days...  
Cleanup complete!  
Files removed: 386  
Empty directories removed: 169  
new-user@new-user-VirtualBox:~$
```